

JONGSEONG BRAD CHOI, Ph.D.

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RESEARCH INTERESTS

Computer Vision; Neural Scene Representation; Physics-Informed AI; Multimodal Spatial AI; Simulation-to-Real Learning; Physical-AI; Human-in-the-loop Robotics; PHM; SHM

EDUCATION

Ph.D., Mechanical Engineering, Purdue University, West Lafayette, IN, USA *06/2020*
- Dissertation: *Automating Visual Data Collection and Analytics toward Lifecycle Management of Engineering Systems*

MSc., Mechanical Engineering, University of Mississippi, University, MS, USA *05/2014*
- Thesis: *Parametric Scramjet Analysis*

BSc., Mechanical Engineering, University of Mississippi, University, MS, USA *08/2012*

EMPLOYMENT HISTORY

Associate Professor (Tenured), Department of Mechanical Engineering .....*08/2026 – Present*
The State University of New York, SUNY Korea, Incheon, South Korea

Secondary Appointment (Courtesy), Department of Computer Science.....*03/2026 – Present*
The State University of New York, SUNY Korea, Incheon, South Korea

Affiliated Faculty (Courtesy), Department of Mechanical Engineering .....*09/2021 – Present*
The State University of New York, Stony Brook University, Stony Brook, NY, USA

Founder & CEO.....*12/2025 – Present*
MEIC AI Co., Ltd., Incheon, South Korea

Assistant Professor, Department of Mechanical Engineering .....*08/2020 – 07/2026*
The State University of New York, SUNY Korea, Incheon, South Korea

Graduate Research Assistant, School of Mechanical Engineering .....*08/2014 – 05/2020*
Purdue University, West Lafayette, IN, USA

Graduate Research and Teaching Assistant, Department of Mechanical Engineering .....*08/2012 – 05/2014*
University of Mississippi, University, MS, USA

Army Corporal, Military Service .....*09/2006 – 08/2008*
First Logistics Support Command, Republic of Korea Army, Inje-Gun, South Korea

PROFESSIONAL POSITION

Associate Editor..... *10/2025 – Present*
ASME Journal of Mechanisms and Robotics, New Jersey, United State

Use Case Provider..... 04*/2026 – Present*
ISO JPEG Committee (Joint Photographic Experts Group), Radiance Field

Committee Chair, Board of Business ..... *01/2025 – Present*

The Korean Society of Mechanical Engineers (KSME) – Division of Reliability Engineering, South Korea
Committee Chair, Board of Internal Business ..... *01/2024 – Present*
The Korean Society of Prognostics and Health Management (PHM Korea), South Korea
Committee, Board of Smart Manufacturing..... *01/2026 – Present*
Korean Society for Precision Engineering (KSPE), South Korea
Committee, Board of Industry-Academia Collaboration ..... *01/2025 – Present*
The Korean Society of Mechanical Engineers (KSME) – Division of IT-Intelligence Convergence, South Korea
Committee, Board of Academic Affairs ..... *01/2025 – Present*
The Korean Reliability Society, South Korea

RESEARCH PROJECTS (Total amount awarded **10,183,265,000 KRW (8,486,054 USD)**; *sole portion*, 1 USD = 1,200 KRW)

40.	Development Program of Titanium Materials for Aircraft Engines (항공기 엔진 국산화 사업), supported by KASA (우주항공청) **(PI) Collaborated with KAI (한국항공우주)** for a research project entitled "*Computer-Vision Engaged Technology Development for Enhanced Structural Integrity of Next-Generation Aerospace Titanium Alloys via Crystallographic Texture Control*"	*7/2027 – 6/2031 (preparing)*
39.	**Research Consultant (PI of MEIC AI Co., Ltd.)**, supported by Korea Water Resources Corporation (K-water), Funding awarded: $458,000 (KRW 549,600,000) for a research project entitled "*Research Service for the Advancement of AI-Based Precision Inspection Technology for Hydraulic Gate Facilities*" **(Lab's spin-off company MEIC-AI won the grant)**	*10/2026 – 03/2029 (ongoing)*
38.	**Research Consultant (PI)**, supported by Doosan Enerbility Co., Ltd., Funding awarded: **$333,333** (KRW 400,000,000) for a research project entitled "*RSDT Agent: Development of an AI Inspection Agent for Detecting Changes in the External Surface and Key Areas of Interest of Reactor Internal Structures during CVAP Inspections*"	*10/2026 – 03/2028 (ongoing)*
37.	**Research Consultant (PI)**, supported by KHNP Central Research Institute (한수원 중앙연), Funding awarded: **$14,667** (KRW 17,600,000) for a research project entitled "*Development of Integrated BOP Design Technologies for Multi-Module Innovative Small Modular Reactors (i-SMRs)*"	*9/2026 – 12/2026 (ongoing)*
36.	**Research Consultant (PI)**, supported by ETRI (한국전자통신연구원). Funding awarded: **$35,833** (KRW 43,000,000) for a research project entitled *"Development of a 3D Gaussian Splatting Generation and Simulation Synthesis Tool Based on Ray Tracing, Integrated with a Next-Generation Neural Network Reconstruction Platform"*	*7/2026 – 11/2026 (ongoing)*
35.	**2026 Strategic Research Program (SRP, 2026 년도 국가전략연구사업) (co-PI)**, supported by NST (국가과학기술연구회). Funding Awarded: requested (sole portion): **$316,667** (380,000,000 KRW) for research entitled "*Development of a Data-Driven Construction-Site-Specific AX Framework and Sustainable Platform (K-CAX)*", Total consortium grant of $32.6M (39.1 billion KRW).	*5/2026 – 12/2030 (ongoing)*
34.	**2026 Future PAN Technology Program (미래 판기술 프로젝트 사업) (co-PI)**, supported by MOTIR (산업통상부). Funding Awarded: **$4,175,000** (5,010,000,000 KRW) for research entitled "*Development of a Real-World–Integrated Spatial Intelligence Platform Based on*	*4/2026 – 12/2033 (ongoing)*

Multimodal 3D Scene Generation and a Superintelligent Operation Engine", Total consortium grant of $20.6M (24.7 billion KRW).

33. **2026 National ICT R&D and Standardization Support Program (2026 년도 제 1 차 정보통신방송 기술개발사업 및 표준개발지원사업) (co-PI)**, supported by IITP(정보통신기획평가원). Funding Awarded: **$416,667** (500,000,000 KRW) for research entitled "*Development of Real–Virtual Synchronization Technology Between Digital Objects for Large-Scale Ultra-Realistic XR Implementation*". Total consortium grant of $4.58M (5.5 billion KRW). *4/2026 – 12/2028 (ongoing)*

32. **2026 National ICT R&D and Standardization Support Program (2026 년도 제 1 차 정보통신방송 기술개발사업 및 표준개발지원사업) (co-PI)** supported by IITP(정보통신기획평가원). Funding Awarded: **$225,000** (270,000,000 KRW) for research entitled "*Multimodal AI–Based Automated Large-Scale Neural Rendering for Sim2Real in Physical AI*". Total consortium grant of $1.50M (1.8billion KRW). *4/2026 – 12/2028 (ongoing)*

31. **Advanced GPU Utilization Support Program (PI)**, supported by NIPA (정보통신산업진흥원). Award of two of B200 GPUs *4/2026 – 5/2026*

30. **Research Consultant (PI)**, supported by ETRI (한국전자통신연구원). Funding awarded: **$41,667 USD** (KRW 50,000,000) for a research project entitled *"Development of a Real-Time Rendering Tool for High-Frequency Detail Restoration via Covariance- and Isotropy-Controlled 3D Gaussian Splatting (Micro-Splatting)."* *10/2025 – 01/2026*

29. **Research Consultant (PI)**, supported by LS Materials and KETI (한국전자기술연구원). Funding awarded: **$22,917 USD** (KRW 27,500,000) for a research project entitled *"Development of Realistic Virtual Environment-Based UC Module Cell Balancing Optimization Technology."* *06/2025 – 09/2025*

28. **Industrial AI and Machine Vision Partnership (PI)**, supported by ITP (인천테크노파크) Funding awarded: **$10,000 USD** (KRW 12,000,000) for a research project entitled *"Machine Vision via Enhanced Industrial AI Platforms and MLOps Implementation."* *05/2025 – 12/2025*

27. **2025 Virtual Convergence Technology Lab. Program (PI)**, supported by RAPA (한국전파진흥협회). Funding awarded: **$568,523 USD** (KRW 682,227,000) for *"MetaReal: Commercialization of Neural Rendering–Based Hyper-Realistic Spatial Information Metaverse Technology."* ***(2025 가상융합기술랩 선정; 수행 4 개대학 – 한국과학기술원, 연세대, 고려대, 한국뉴욕주립대)*** *05/2025 – 12/2027 (ongoing)*

26. **2025 Hyundai Motor Company Future Technology Research Program (PI)**, supported by Hyundai Motor Group (현대자동차그룹). Funding awarded: **$55,000 USD** (KRW 66,000,000) for a research project entitled *"Physics-Informed Neural Network (PINN)-Based EV Energy Consumption PHM Model and EV Navigation System Development."* ***(2025 현대자동차 미래기술공모과제 선정)*** *05/2025 – 12/2025*

25. **2025 Excellent Young Researcher Program (우수신진연구) (PI)**, supported by NRF (한국연구재단 우수신진연구). Funding awarded: **$1,006,541 USD** (KRW 1,207,849,000) for a research project entitled *"True Twin: AI-based Hyper-Realistic Industrial Digital Twin for Distributed Collaborative and Predictive Facility Maintenance in Mixed Reality."* ***(2025 한국연구재단 우수신진연구 선정; equivalent to NSF CAREER)*** *03/2025 – 02/2030 (ongoing)*

24. **Research Consultant (PI)**, supported by KIMM (한국기계연구원) and Doosan Enerbility (두산에너빌리티). Funding awarded: **$15,191 USD** (KRW 18,229,000) for a research project entitled *"Development of High-Resolution Modeling for Nuclear Reactor and Internal Structures with Neural Rendering."* — *01/2025 – 06/2025*

23 **Graduate Student Support Partnership (Collaborator)**, supported by Sungkyunkwan University. Funding awarded: **$20,000 USD** (KRW 24,000,000) for the project entitled "*AI-Driven Reliability and Thermal Optimization for Semiconductor Packages***.**" — *01/2025 – 03/2025*

22. **Promising Labs Graduate Student Research Support Program (PI, Internal)**, supported by SUNY Korea R&BDF. Funding awarded: **$14,667 USD** (KRW 17,600,000). — *09/2024 – 12/2024*

21. **International Joint Research Support Program (PI, Internal)**, supported by SUNY Korea R&BDF. Funding awarded: **$7,500 USD** (KRW 9,000,000). — *06/2024 – 11/2024*

20. **Research Consultant (PI)**, supported by KR (한국선급). Funding awarded: **$36,667 USD** (KRW 44,000,000) for a research project entitled "*Remote Ship Inspection online Platform*." — *03/2024 – 10/2024*

19. **Research Consultant (PI)**, supported by MDS Intelligence Inc (MDS 인텔리전스). Funding awarded: **$50,417 USD** (KRW 60,500,000) for a research project entitled "*Indoor 3D Modeling for Facility Digital Twins***.**" — *11/2023 – 2/2024*

18. **2023 Global Product Service Localization Research (PI)**, supported by ITP (인천테크노파크). Funding awarded: **$4,167 USD** (KRW 5,000,000). — *08/2023 – 10/2023*

17. **Research Consultant (PI)**, supported by KR (한국선급). Funding awarded: **$36,667 USD** (KRW 44,000,000) for a research project entitled "*Robot Vision-Based Ship Engine Sensing***.**" — *05/2023 – 12/2023*

16. **STEAM Activity Research Program for Gifted High School Students (PI)**, supported by IASA (인천과학예술영재학교). Funding awarded: **$13,333 USD** (KRW 16,000,000). — *03/2023 – 12/2023*

15. **2022 Global Product Service Localization Research (PI)**, supported by ITP (인천테크노파크). Funding awarded: **$4,167 USD** (KRW 5,000,000). — *09/2022 – 12/2022*

14. **Basic Research Program (PI)**, supported by NRF (한국연구재단). Funding awarded: **$84,600 USD** (KRW 101,520,000) for a research project entitled *"Emulation and Enhancement of Human Capability on Infrastructure Assessment: Engineering-Soft-Power Engaged Monitoring and 3 Major Techniques Development."* — *06/2022 – 02/2024*

13. **Industry-Academia Research Consortium for Future Mobility (Co-PI)**, supported by ITP (인천테크노파크). Funding awarded: **$23,333 USD** (KRW 28,000,000; 46.67% share of total $50,000) for a research project entitled "*Smart-Car MCU Component and Module Development***.**" — *05/2022 – 12/2022*

12. **Space Exploration and In-Situ Resource Utilization (ISRU) Core Technology Research Center Program (Co-PI)**, supported by KASA (우주항공청). Funding awarded: **$395,833 USD** (KRW 475,000,000; 4.03% share of total $9.83M) for a research project entitled *"Space Exploration and In-Situ Resource Utilization Center (SRC)."* — *04/2022 – 12/2026 (ongoing)*

11. **Research Consultant (PI)**, supported by AIDL Co. Funding awarded: **$4,750 USD** (KRW 5,700,000) for a research project entitled "*IMU-Based Autonomous Guided Vehicle Control System***.**" — *03/2022 – 07/2022*

10. **2021 Global Product Service Localization Research (PI)**, supported by ITP (인천테크노파크). Funding awarded: **$8,333 USD** (KRW 10,000,000). *10/2021– 12/2021*

9. **Equipment Grant (PI)**, supported by KICT (한국건설기술연구원). **A**ward of 3 UGVs and Velodyne LiDAR Puck equivalent to **$44,167 USD** (KRW 53,000,000) *06/2021 – 06/2031 (ongoing)*

8. **2021 Metaverse Lab. (Co-PI)**, supported by RAPA (한국전파진흥협회). Funding awarded: **$17,250 USD** (KRW 20,700,000; 3.45% share of total $500,000) for a project entitled "*XR-Based Intuitive Inventory System for Smart Factory***.**" *04/2021 – 12/2021*

7. **First Career Research Grant (PI)**, supported by NRF (한국연구재단). Funding awarded: **$25,200 USD** (KRW 30,240,000) for a research project entitled "*Automating Visual Assessment of Infrastructure Exploiting Computer Vision and Big Visual Data*". *03/2021 – 02/2022*

6. **NSF OAC (Office of Advanced Cyberinfrastructure) (Graduate RA)** (Grant No. NSF-1835473) for a research project entitled "*Integrating Human and Machine for Post-Disaster Visual Analytics***.**" *01/2019 – 05/2020*

5. **EU Horizon 2020 Research & Innovation Programme (Graduate RA)** (Grant No. 700191) for a research project entitled **"***STORM: Cultural Heritage Risk Management*." *04/2017 – 05/2020*

4. **NASA Space Technology Mission Directorate (STMD) (Graduate RA)** for a research project entitled "*Resilient ExtraTerrestrial Habitats Institute (RETHi / RETH)*." *08/2018 – 05/2020*

3. **NSF CAREER Program (Graduate RA)** (Grant No. NSF-1645047) for a research project entitled "*Active Citizen Engagement for Infrastructure Lifecycle Management*." *05/2017 – 08/2018*

2. **State Planning and Research (SPR) Program (Graduate RA)**, supported by INDOT (Grant No. SPR-4006) for a research project entitled "*Image-Based Collection and Measurement for Construction Pay Items.***"** *05/2015 – 08/2017*

1. **Graduate Student Support Awards (Graduate RA)**, supported by the University of Mississippi for a research entitled "*Parametric Analysis of Scramjet Engine Performance*." *08/2012 – 05/2014*

TEACHING RECORDS

Course Teaching

MEC 559: (Graduate Course) Mobile Robotics and Autonomous Vehicle – Computer Vision and Control at *the State University of New York, SUNY Korea* *Sp 24 – 26*

MEC 510: (Graduate Course) Object-Oriented Programming for Scientists and Engineers – Visual Analytics for Mechanical Engineers at *the State University of New York, SUNY Korea* *Sp22 – 26*

MEC 410: Design of Machine Elements at *the State University of New York, SUNY Korea* *Sp22 – 23*

MEC 301: Thermodynamics at *the State University of New York, SUNY Korea*: Recorded the highest course evaluation score in the department *Fa20 – 26*

MEC 320: Numerical Methods in Engineering Design and Analysis at *the State University of New York, SUNY Korea* *Fa21 – 25*

MEC 363: Mechanics of Materials at *the State University of New York, SUNY Korea* *Sp21*

STUDENT SUPERVISED (Total of **5 Ph.D.**, **16 Master's**, **44 bachelor's** level RAs)

RAs – Course of Ph.D., *the State University of New York, SUNY Korea*

1. **(Graduated**) Jonathan Boyack: Engineering-Soft-Power: A New Engineering Approach through Visual Intelligence, Machine Learning, and Hyper-Realistic Digital Twin **(appointed postdoc at Purdue Univ. IN, USA)** *2021.03 – 2025.12*
2. Ricardo Ortiz: Development of RKTGPS-based Autonomous Vehicle using ROS (LoA) *2024.03 – Present*
3. Hansol Lim: Physics-Informed Neural Operator for Sim2Real in Physical-AI *2024.09 – Present*
4. Alfredo Valenzuela: An Automated Camera Calibration for Long-distance Assessment *2025.03 – Present*
5. Jee Won Lee: MicroSplats: Image-based 3D Reconstruction using Micro Gaussian Splatting *2025.09 – Present*

RAs – Course of Master's, *the State University of New York, SUNY Korea*

1. **(Graduated**) Hanbeom Chang: SRCX: Advanced Visual Sensor Fusion for Remote Structural Assessment and Collaborative Inspection in Extended Reality **(currently at SK AX Co., Ltd., South Korea)** *2023.03 – 2025.06*
2. **(Graduated**) Jee Won Lee: A Unified Framework for High-Fidelity 3D Gaussian Splatting and Remote Inspection from Micro-splatting to ReVIEW (currently in a Ph.D. program, Research Assistant, SUNY Korea) *2022.03 – 2025.06*
3. **(Graduated**) Alfredo Valenzuela: An Automated Long-Range Pan-Tilt-Zoom Target Control and Mapping App System for Remote Sensing (currently in a Ph.D. program, Research Assistant, SUNY Korea) *2022.08 – 2024.12*
4. **(Graduated**) Hansol Lim: LiDAR-3DGS: LiDAR Reinforced 3D Gaussian Splatting for Accurate Digital Twins (currently in a Ph.D. program, Research Assistant, SUNY Korea) *2022.08 – 2024.06*
5. **(Graduated**) Ricardo Ortiz: Autonomous Manipulation of Unmanned Ground Vehicle (UGV) with an User-Guided-Path on a Map: Quantitative Simulation and Analysis (currently in a Ph.D. program, Research Assistant, SUNY Korea) *2021.08 – 2023.12*
6. **(Graduated**) Jonathan Boyack: Non-Thesis option (currently in a Ph.D. program, Research Assistant, SUNY Korea) *2020.08 – 2021.12*
7. Hyeoji Chang: 3D Gaussian Splatting SLAM (LoA) *2025.03 – Present*
8. Sooyeun Yang: Digital Twin Operation through Issac SIM *2025.03 – Present*
9. Minhyeok Im (CS major): Neural Rendering engaged SLAM *2025.09 – Present*
10. Nahyun Kim (CS major, engaged): Sim2Real in Physical-AI engaged Smart Factory *2026.03 – Present*
11. Cheyul Im: Hyper-Resolution 3D Reconstruction and Blender *2026.03 – Present*
12. Sungwook Choi: Multimodal-AI through Mixed/Virtual Reality *2026.03 – Present*
13. Sunghoon Jeong: Neural Rendering based Sim2Real & Physical-AI *2026.03 – Present*
14. Semin Bae: Real-time and Spatial Synchronization for Teleoperation System *2026.03 – Present*
15. Hugo Zuniga: Imitation Learning for Robot Foundation Model *2026.09 – Present*
16. Felipe Zuniga: Robot Simulation for Simulation-to-Real Problems *2026.09 – Present*

RAs – Funded Undergraduate Researchers, *the State University of New York, SUNY Korea*

1. **(Graduated)** Cheyul Im (master's program at MEC at SUNY Korea) *2025.03 – 2025.12*
2. **(Graduated)** Sungwook Choi (master's program at MEC at SUNY Korea) *2025.03 – 2025.12*
3. **(Graduated)** Sunghoon Jeong (master's program at MEC at SUNY Korea) *2025.03 – 2025.12*
4. **(Graduated)** DaeRyeon Choi **(master's program at Univ. of Michigan Ann Arbor)** *2024.01 – 2024.12*
5. **(Graduated)** Mijin Lee (GM Korea, South Korea) *2024.01 – 2024.12*
6. **(Graduated)** Jungmin Lee **(Hyundai Mobis, USA)** *2024.01 – 2024.12*
7. **(Graduated)** Eunjae Lee (Skyes Blackchain, Co., South Korea) *2024.01 – 2024.12*
8. **(Graduated)** Soo-Jung Chi **(Johnson & Johnson Innovative Medicine, South Korea)** *2024.01 – 2024.12*
9. **(Graduated)** Hojin Song (Army, Military Service, South Korea) *2024.01 – 2024.12*
10. **(Graduated)** Sooyeun Yang (master's program at MEC, SUNY Korea) *2024.01 – 2024.12*

11. **(Graduated)** Hyobin Shin (running own business, South Korea) *2024.01 – 2024.12*
12. **(Graduated)** Leeloy Makusha **(master's program at UT Alington, USA)** *2023.01 – 2023.12*
13. **(Graduated)** Yoonseong Kim (Research Intern at KITECH, South Korea) *2023.07 – 2023.12*
14. **(Graduated)** Sambridha Bhattarai (owner of Cinder Bar, South Korea) *2023.07 – 2023.12*
15. **(Graduated)** Dannielle Macmaster (an engineering Company, USA) *2023.01 – 2023.12*
16. **(Graduated)** Hanbeom Chang (master's program at MEC at SUNY Korea) *2022.07 – 2023.06*
17. **(Graduated)** Jaewon Lee (master's program at MEC at SUNY Korea) *2022.07 – 2023.06*
18. **(Graduated)** Yijoo Um **(Korea Aerospace Industries - KAI, Ltd., South Korea)** *2022.07 – 2023.06*
19. **(Graduated)** Seunghyun Cha (KT, Co., Ltd., South Korea) *2022.01 – 2024.12*
20. **(Graduated)** Jimin Shin (master's program at MEC at SUNY Korea) *2022.01 – 2024.12*
21. **(Graduated)** Pureun Jeong (Yujin Robotics, Co, Ltd., South Korea) *2021.07 – 2022.06*
22. **(Graduated)** Hojung Lim **(master's program at CS at USC, USA)** *2021.07 – 2021.12*
23. **(Graduated)** Hyeji Chang **(undergrad program at Georgia Tech, USA)** *2020.07 – 2021.06*
24. **(Graduated)** Jeewon Lee (master's program at SUNY Korea, South Korea) *2020.07 – 2021.12*
25. **(Graduated)** Hansol Lim (master's program at SUNY Korea, South Korea) *2020.07 – 2021.03*
26. **(Graduated)** Sooyon Chang **(master program at Cornell University, USA)** *2020.07 – 2021.12*
27. **(Graduated)** Hyunseung Cha **(Hyundai Rotem, Co., Ltd., South Korea)** *2020.07 – 2021.12*
28. **(Graduate)** Gunwoo Kim: Camera Calibration for Rapid Projection Matrix Generation *2025.01 – 2025.13*
29. **(Graduated)** Prince-David Malendele **(Brookhaven Natl. Lab., USA)** *2020.07 – 2021.12*
30. Justin Kim: Neural Rendering and Multimodal AI *2026.01 – Present*
31. Andrew Kim: AGI and Safe AI *2026.01 – Present*
32. Jaesuk Song: XR-based teleoperations *2026.01 – Present*
33. Janheum Park: Voxel-based 3D reconstruction *2026.01 – Present*
34. Jeonghwan Jeikal: Physics-informed Neural Operator *2026.01 – Present*
35. Christopher Song: Physics-informed Neural Operator *2026.01 – Present*
36. Seungho Jun (CS major): Front-End App Development for Real-time Rendering *2025.07 – Present*
37. Dohyun Lee (CS major): Voxel based 3D Reconstruction *2025.07 – Present*
38. Sanha Jang: Neural Rendering based Hyper-Realistic Spatial Information Metaverse *2025.07 – Present*
39. Hajun Jang: Neural Rendering based Hyper-Realistic Spatial Information Metaverse *2025.07 – Present*
40. Eunha Choi: Neural Rendering based Hyper-Realistic Spatial Information Metaverse *2025.07 – Present*

Research Intern via International MOU

41. Lesus Aguilla (Research Intern from Ecuador): UGV Swarm: inter-connected through LiDAR and ROS, Undergraduate student, Dept. of Mechatronics, Escuela Superior Politécnica del Litoral, Ecuador *2023.01 – 2023.05*
42. Hugo Zuniga (Research Intern from Ecuador): Adaptive Control of PTZ sensor Undergraduate student, Dept. of Mechatronics, Escuela Superior Politécnica del Litoral, Ecuador *2024.01 – 2024.05*
43. Felipe Zuniga (Research Intern from Ecuador): Adaptive Control of 7-DOF Robot Arm Undergraduate student, Dept. of Mechatronics, Escuela Superior Politécnica del Litoral, Ecuador *2024.01 – 2024.05*

Dissertation Committee, **Ph.D. Student** other than my own students**,** *the State University of New York, SUNY Korea*

1. Mark Anthony Rotor: Designing tidal turbine (HATT) blades utilizing Artificial Neural Network (ANN) where I serve as chair of the committee (Supervisor Prof. Hefazi) *2021.01– 2023.08*

2. Saebom Jin: Breaking the Vicious Cycle: How Vulnerable Cities Can Develop Smart and Sustainable with Adaptation Planning (Supervisor Prof. Maliphol) *2023.04– 2023.12*

Thesis Committee, **Master Student** other than my own students**,** *the State University of New York, SUNY Korea*

1. Suyeon Lee: Prognostics using Nonlinear Cumulative Damage Model for Electronic Devices where I serve as member of the committee (Supervisor Prof. Han) *2021.08– 2023.06*

Research Mentor, **Undergraduate Research Course**, *Purdue University*

- Wookjin Chung: 6 credits of undergraduate research (Sp2018, Fal2019)
- Jonghyun Park: 6 credits of undergraduate research (Sp2018, Fa2019)
- Gun Wook Park: 6 credits of undergraduate research (Sp2017, Fa2017)
- Sharda Parth: 3 credits of undergraduate research (Sp2018)
- Yisong Yin: 6 credits of undergraduate research (Fa2016, Sp2017)

VISITING SCHOLARS

1. **Prof. Chul Min Yeum** from University of Waterloo, ON, Canada: Teleoperation through XR platform *2025.10 – 2026.09*
2. **Prof. Francisco Yumbla** from Escuela Superior Politécnica del Litoral (ESPOL), Ecuador: 7DOF Robot Arm for High-Precision Image Sensor Zoom Control *2025.10 – 2026.01*

CURRICULUM DEVELOPMENT AND EDUCATIONAL SERVICE

Graduate Course Development – *MEC 559 Mobile Robotics and Autonomous Vehicles: Computer Vision and Control* *2024 - present*

- Students are to be trained to manipulate a Robot Operating System (ROS) package, work on related tasks

Graduate Course Development – *MEC 510 Object-Oriented Programming for Scientists and Engineers: Visual Analytics and Sensing* *2022 – present*

- Providing Computer Vision (CV) and Machine Learning (ML) techniques in various mechanical engineering subjects

High School Credit System and Curriculum *2016 – 2017*

- Chosen and served as an instructor for the 16-hour course “Computer Vision and Sustainable Cities Applications” organized by Incheon of Education

TRAILS: Teachers and Researchers Advancing Integrated Lessons in STEM, *Purdue University* *2016 – 2017*

- Supported by NSF under Grant No. NSF-1513248
- Participated as a graduate research assistant in promoting practices that increase students' motivations and capacities to pursue careers in STEM area.

SLED: Science Learning through Engineering Design, *Purdue University* *2015- 2016*

- Supported by NSF under Grant No. NSF-0962840
- Participated as a graduate research assistant in collaboration between STEM disciplinary faculty and grades 3-6 teachers.

HONORS & AWARDS (25 Recognitions/ Achievement / Award) *** corresponding author or PI**

Awards

1. **Best Paper Award** from Intl. Conf. on Quality, Reliability, Risk, Maintenance, and Safety Engineering (QR2MSE 2025, Hohhot, China) *07/2026*
   - **Jongseong Choi***, "Visual Intelligence for Predictive Engineering System Assessment"
2. **Outstanding Presentation Award** from Intl. Conf. on Precision Engineering and Sustainable Manufacturing (PRESM 2025, Chiangmai, Thailand) *07/2025*
   - **Jongseong Choi***, "Physics-informed Neural Network (PINN) and Hyper-realistic Digital Twin (3DGS) for Predictive Engineering System Assessment"
3. **Promising Young Scientist Award** from Korean Society of Mechanical Engineers Reliability Division 2025 (KSME Reliability 2025, Gyeongju, South Korea) *04/2025*
   - **Jongseong Choi***, "EV-PINN: Electric Vehicle Battery Surrogate Model using Transformer-Based Physics Informed Neural Network"
4. **Grand Prize** from "Idea Challenge Competition" held by Korean Society of Mechanical Engineers Chungcheong Division and KIMM 2024 (KSME Chungcheong 2024, Daejeon, South Korea) *12/2024*
   - Hansol Lim, Jeewon Lee, Hanbeom Chang, Sooyeun Yang, and **Jongseong Choi***, "Physics-Informed Neural Network for EV Battery Prediction"
5. **Outstanding Paper Award** from 7th Intl. Conf. on Materials and Reliability (ICMR 2025, Busan, South Korea) *12/2024*
   - Jeewon Lee, Hansol Lim, **Jongseong Choi***, Sangho Song, Jaechul Park, and Hwasup Jang, "Development of a 3D Gaussian Splatting Digital Twining Platform for Remote Ship Assessment"
6. **Vice Presidential Award for Industrial Potential** from 2024 SUNY Korea Research Challenge *12/2024*
   - The team of Hansol Lim, Jeewon Lee, Hanbeom Chang, and **Jongseong Choi*** won the competition and a monetary prize of $10,000 for conference attending
7. **Best Poster Award** from Annual Conf. of the Korean Society of PHM (PHM Korea 2024, Seoul, South Korea) *10/2024*
   - Hanbeom Chang and **Jongseong Choi***, "LiDAR-Camera based Real-time 3D Reconstruction with Mixed Reality Visualization System"
8. **Best Paper Award** from Korean Society of Mechanical Engineers Reliability Division 2024 (KSME Reliability 2024, Yeosu, South Korea) *03/2024*
   - Jonathan Boyack and **Jongseong Choi***, "Falling Hazard Assessment Application: Automated Labelling Large Data Sets for Transfer Learning Models"
9. **Outstanding Thesis Award** from Korean Society of Mechanical Engineers Reliability Division 2024 (KSME Reliability 2024, Yeosu, South Korea) *03/2024*
   - Ricardo Ortiz and **Jongseong Choi**, "Autonomous Manipulation of Unmanned Ground Vehicle (UGV) with an User-Guided-Path on a Map: Quantitative Simulation and Analysis"
10. **Outstanding Paper Award** from 14th Intl. Symposium on NDT in Aerospace (AeroNDT 2023, Busan, South Korea) *11/2023*
    - Hansol Lim and **Jongseong Choi***, "Real-Time 3D Video Generation Using NeRF"
11. **Best Paper Award** from 14th Intl. Symposium on NDT in Aerospace (AeroNDT 2023, Busan, South Korea) *11/2023*
    - Jonathan Boyack, Hansol Lim, Alfredo Valenzuela, **Jongseong Choi***, Sangho Song, Yongseok Choi, and Dongguk Im, "Region of Interest Locator with Digital Twin Model Generation of Large-Scale Facility"

12. **Excellent Presentation Award** from Korean Society of Mechanical Engineers Annual Meeting 2023 (KSME 2023, Incheon, South Korea) *11/2023*
    - Hansol Lim and **Jongseong Choi***, “Seamless 3D Scene Control of NeRF Generated Digital Twin Environments”
13. **Best Paper Award** from Korean Society of Mechanical Engineers Annual Meeting 2023 (KSME 2023, Incheon, South Korea) *11/2023*
    - Jonathan Boyack and **Jongseong Choi***, “Falling Hazard Assessment Application: Automated Labelling Large Data Sets for Transfer Learning Models”
14. **Bronze Medal** from Korean Society of Mechanical Engineers Annual Meeting 2023 (KSME 2023, Incheon, South Korea) *11/2023*
    - Hansol Lim participated a LLM Competition and won with title “ LLM-based Fault Diagnosis and Natural Language Response in Military Robots and Drones”
15. **Outstanding Project Award** from Incheon Technopark (ITP) *11/2023*
    - The team MEIC (Ricardo Ortiz, Jonathan Boyakck, Alfredo Valenzuela, and Sambridha Bhattarai, supervised by **Jongseong Choi**, won "2023 Global Product-Service Localization Research Award Competition”
16. **Promising Young Scientist Award** from Annual Conference of the PHM Korea Society (PHM Korea 2023, Seoul, South Korea) *07/2023*
    - **Jongseong Choi***, "Engineering-Soft-Power engaged visualization analytics and human-machine collaboration remote assessment for large-scale structures"
17. **Best Paper Award** from Annual Conference of the PHM Korea Society (PHM Korea 2023, Seoul, South Korea) *07/2023*
    - Hansol Lim and **Jongseong Choi***, “Real-Time, Non-Intrusive Inspection of Structures and Machines Using Visual Analytics and SfM-Based Digital Twin Generation”
18. **Best Paper Award** from Korean Society of Mechanical Engineering Annual Meeting 2022 (KSME 2024, Jeju, South Korea) *11/2022*
    - Jonathan Boyack and **Jongseong Choi***, "A Real-Time Log Data Analytics-based Electric Vehicle Path Optimization", only 2 papers were selected for this award among all papers submitted to the Reliability Division in KSME.
19. **Best Paper Award** from Korean Society of Mechanical Engineering Annual Meeting 2022 (KSME 2024, Jeju, South Korea) *11/2022*
    - Pureun Jeong, Jimin Shin, Seung Hyun Cha, **Jongseong Choi***, "Indoor Navigation System Designed for a Safety of Construction Site", Only 2 papers were selected for this award among all papers submitted to the Reliability Division in KSME.
20. **Outstanding Project Award** from Incheon Technopark (ITP), Incheon, South Korea *11/2022*
    - The team MEIC (Jonathan Boyakck, Ricardo Ortiz, and Jee Won Lee, supervised by Prof. Jongseong Choi, won "2022 Global Product-Service Localization Research Award Competition”
21. **Travel Award & Workshop Invitation** from NHERI RAPID Experimental Facility, NSF (UW Seattle, WA, USA) *07/2019*
    - This award selects 20 attendees who has professional research background for the RAPID 4 days Equipment Training Workshop at the University of Washington, Seattle, as well as support up to $1,500 Travel Fund.
22. **Travel Award for Conference** from College of Engineering, Purdue University *05/2018*

23. **Honorable Mentions** from 3rd Midwest Smart Structure Colloquium, University of Illinois (UIUC) *10/2017*
    - This award recognizes excellence PhD candidates supporting up to $1,000 for 2019 EWSHM conference at Hilton Hotel, Manchester, UK
24. **Resident Assistant Scholarship** from University of Mississippi *08/2012 – 08/2013*
25. **Honor Program Scholarship** from University of Mississippi *01/2011*
    - This scholarship is awarded to prominent undergraduate students in the School of Engineering.

SYNERGISTIC LEADERSHIP POSITION

Organizing Member of International Conference

- Organizing committee member of the 8th International Conference on Materials and Reliability and 8th International Conference (QR2MSE 2025) on Materials and Reliability (ICMR 2025) *07/2025*
- Organizing committee member of the 7th International Conference on Materials and Reliability (ICMR 2024) *11/2024*
- Steering committee member of the 14th International Symposium on NDT in Aerospace (AeroNDT 2023) *11/2023*

Organizing Member of Korean Conference

- Committee member for the Annual Spring Conference of the Korean Society of Mechanical Engineers (KSME) - Reliability Division *2023 – 2026*
- Committee Member for the Annual Conference of PHM Korea *2022 – 2025*

Membership in Professional Society

- Member of American Society of Mechanical Engineers (ASME) *2012 – 2014*
- Member of the American Institute of Aeronautics and Astronautics (AIAA) *2014 – 2015*
- Member of the Institute of Electrical and Electronics Engineers (IEEE) / Robotics and Automation Society (RAS) *2017 – Present*
- Member of the Korean Society of Prognostics and Health Management (KSPHM) *2021 - Present*
- Member of Korean Society of Mechanical Engineers (KSME) - Reliability Division *2022 - Present*
- Member of Korean Society of Mechanical Engineers (KSME) – IT-Intelligence Convergence *2025 - Present*
- Member of The Korean Reliability Society (KORAS) *2025 - Present*
- Member of Korean Computer Vision Society (KCVS) *2025 - Present*

Expert Position

- External Advisory Board, OnePredict Co., Ltd. *2026 – Present*
- Member of the Proposal Evaluation Committee, 5th Term, Gyeonggi Housing Corporation *2025 – Present*
- Korea Metaverse Talent Development Committee *2025 – Present*

- Member of the Project Evaluation/Selection Committee for National Research Foundation of Korea (NRF), Ministry of Trade, Industry and Energy (MOTIE), Ministry of Land, Infrastructure and Transport (MOLIT), Ministry of Science and ICT (MSIT), Incheon Technopark (ITP), and many *2020 – Present*

PROFESSIONAL TALKS & PRESENTATION

[T34] Invited Lecturer, "How Physical AI Learns About the World" Sebasi Lecture Series, CBS Broadcasting, South Korea *10/2026*

[T33] Invited Seminar, "Micro-Splatting: Bridging the Sim2Real Gap for Physical AI and Robot Foundation Models", Sungkyunkwan University, South Korea *10/2026*

[T32] Invited Seminar, "Robot Foundation Model (RFM), Sim2Real, and Neural Rendering", Korea Institute of Machinery and Materials (KIMM), Daejeon South Korea *09/2026*

[T31] Invited Seminar, "Simultaneous Spatial Twinning and Teleoperation Technologies for Defense Applications Based on Spatial AI and Multimodal AI", LIG Nex1, LIG Nex1 Pangyo Office, Seongnam, South Korea *05/2026*

[T30] Invited Speaker, "Sim2Real and Neural Rendering: Next-Generation BIM Based on Visual Intelligence", Scan to BIM with AI Seminar Series, Korea BIM Association, Online *04/2026*

[T29] Guest Lecturer, "DT-based simulation for manufacturing Physical-AI", Manufacturing Physical-AI Curriculum Lecture Series held by Advanced AI Manufacturing Innovation Professional Workforce Development Program, Seoul National University, L7 Hotel, Seoul, South Korea *02/2026*

[T28] Guest Lecturer, "Physical-AI and Fundamentals: Sim2Real through Neural Rendering and Visual Intelligence", Professional Training Course Series entitled "PHM Technologies Utilizing Signal Processing, Agentic AI, and Physical AI", Hanyang University, Seoul, South Korea *02/2026*

[T27] Invited Speaker, "Digital Twin-Based Structural Anomaly Detection Technology Using Hyper-Realistic Visual Intelligence", Workshop on Advanced Reactor System Design Technologies for Practical Research Applications, hosted by Reactor Systems Engineering Division of Korean Nuclear Society (KNS), Changwon Convention Center, Changwon, South Korea *10/2025*

[T26] Invited Speaker, "Digital Twin–Driven Structural Anomaly Detection Using Hyper-Realistic Visual Intelligence Neural Rendering", hosted by Korea Aerospace Industries, Ltd. (KAI), KAI Headquarter, Sacheon, South Korea *10/2025*

[T25] Invited Speaker, "Digital Twin–Based Structural Anomaly Detection Technology Utilizing Hyper-Realistic Visual Intelligence and Neural Rendering", Nuclear and Energy Division Workshop, hosted by Korean Society for Noise and Vibration Engineering (KSNVE), Atomic Energy Research Institute (KAERI), Daejeon, South Korea *09/2025*

[T24] Invited Speaker, "MEIC Lab's Ongoing Research Introduction: Physical AI current and Future", Global AI Researcher Forum, hosted by AI Research Center in K-water, K-water Research Center, Daejeon, South Korea *09/2025*

[T23] Invited Speaker, "Hyper-Realistic Digital Twin and 3D Perception for Predictive Engineering System Assessment", Graduate Seminar Series, hosted by Dept. of Math, Ewha Women's University, Seoul, South Korea *09/2025*

[T22] Invited Speaker, “Hyper-Realistic AI current and future”, Workshop of Autonomous Vehicle & Robotics Division, hosted by Institute of Control, Robotics and Systems (ICROS), KAIST Dongtan, Osan, South Korea *08/2025*

[T21] Invited Speaker, “Digital Twin-based Structural Anomaly Detection Technology Utilizing Visual Intelligence”, Workshop on Advanced SMR Design Technologies, hosted by Central Research Institute of Korea Hydro & Nuclear Power (KHNP), Koreana Hotel, Seoul, South Korea *08/2025*

[T20] Invited Speaker, “Hyper-realistic 3D Neural Rendering Technology and Physics-Informed Neural Networks for Intelligent Industrial Equipment Prognostics and Health Management”, Central Research Institute Seminar Series, hosted by Korea Hydro & Nuclear Power (KHNP), KHNP, Daejeon, South Korea (Jul 25) *07/2025*

[T19] Invited Lecturer, “Physics-Informed Neural Network; History and Current”, Workshop on Industrial Application Technologies in the Energy and Machinery Sector for Carbon Neutrality, hosted by Korea Refrigeration and Air conditioning Assessment Center (KRAAC), SUNY Korea, Incheon, South Korea *01/2025*

[T18] Invited Speaker, “Human-Machine Distributed Collaborative Remote Sensing and Visual Analytics for Asset Assessment”, ME Graduate School Seminar Series, hosted by AI-PHM Lab., Sungkyunkwan University, Suwon, South Korea *01/2025*

[T17] Invited Speaker, “Visual Analytics; History and Current”, Workshop of Autonomous Vehicle & Robotics Division, hosted by Institute of Control, Robotics and Systems (ICROS), Holiday Inn Songdo, Incheon, South Korea *10/2024*

[T16] Invited Speaker, “Human-Machine Distributed Collaborative Remote Sensing and Visual Analytics for Asset Assessment”, ME Graduate School Seminar Series, hosted by Dept. Mechanical Engineering, UNIST, Ulsan, South Korea *10/2024*

[T15] Invited Plenary Speaker, “Visual Analytics; History and Current”, Inha University 70th anniversary symposium, hosted by Inha University, Incheon, South Korea *04/2024*

[T14] Invited Speaker, “Remote Sensing through Engineering-Soft-Power”, Invited Seminar Series, hosted by Korea Electronics Technology Institute (KETI), Bucheon, South Korea *12/2023*

[T13] Invited Speaker, "A Real-Time Log Data Analytics-based Electric Vehicle Path Optimization", Expert Seminar Series, hosted by Vehicle & Mobility Systems Department (VMS), Argonne National Lab., Lemont, IL, USA *05/2023*

[T12] Invited Guest Lecturer, “Human-Machine Collaborative Remote Monitoring and Sensing”, Expert Seminar Series, hosted by Digital Twin Team, Korean Register, Busan, South Korea *12/2022*

[T11] Invited Guest Lecturer, “Human-Machine Collaborative Remote Monitoring and Sensing”, ME Seminar Series, hosted by Mechanical Engineering, Hanyang University, Seoul, South Korea *11/2022*

[T10] Invited Guest Lecturer, “Computer Vision engaged Infrastructure Remote Sensing”, Dept. Architecture Engineering Seminar Series, hosted by Architecture Engineering, Inha University, Incheon, South Korea *11/2022*

[T9] Invited Speaker, “Citizen Engaged Infrastructure Assessment and Remote Sensing”, Young Promising Scientists Session, hosted by Korean Society of Mechanical Engineers (KSME) Annual Conference 2022, Seoguipo, South Korea *11/2022*

[T8] Invited Guest Lecturer, “Cultural Heritage Long-term Preservation and Monitoring”, Expert Seminar Series, hosted by National Museum of Modern and Contemporary Art, Cheongju, South *09/2022*

[T7] Invited Guest Lecturer, "Citizen Engaged Infrastructure Assessment and Remote Monitoring", Artificial Intelligence Winter School, hosted by AI-Machine Research Group in The Korean Society for Mechanical Engineers (KSME), KAIST, Daejeon, South Korea *02/2022*

[T6] Research Seminar, "AI to Enable Water Quality Control", Expert Seminar Series, hosted by Magpiesoft co., Daejeon, South Korea (Feb *02/2022*

[T5] Research Seminar, "AI for Possible Military Applications", Expert Seminar Series, hosted by HANCOM Intelligence co., Seongnam, South Korea *11/2021*

[T4] Research Seminar, "Citizen Engaged Building Cultural Heritage Management", Expert Seminar Series, hosted by Cultural Heritage Department of Incheon City, Incheon City Hall, Incheon, South Korea (*11/2021*

[T3] Invited Speaker, "Automating Visual Data Analytics to Aid Lifecycle Management of Infrastructure toward Sustainable Cities", Conference on Toward 'Ontact' Industries through PHM (PHM Korea 2021), hosted by Korea Society for Prognostics and Health Monitoring (KSPHM), Jeju, South Korea *09/2021*

[T2] Invited Seminar, "Infrastructure Management with Automated Vision Systems", Expert Seminar Series, hosted by Incheon City Museum, Incheon, South Korea *10/2021*

[T1] Invited Seminar, "Vision-based visual inspection for large-scale infrastructure", Expert Seminar Series, hosted by Korea Institute of Construction Technology (KICT), Ilsan, South Korea *09/2020*

PATENTS (1 Registered; 8 Filed) *** Inventor**

[A9] *METHOD OF ESTIMATING DYNAMIC COEFFICIENT OF ELECTRIC VEHICLE AND PREDICTING ENERGY CONSUMPTION BY UTILIZING SPEED LOG DATA AND PHYSICS-INFORMED NEURAL NETWORK,* **Jongseong Choi***, filed at the **United States** in Aug 2025 (19/290,373)

[A8] *COMPUTER VISION TECHNIQUE TO EXTRACT STRUCTURAL INFORMATION FROM CITIZEN SCIENCE DATA: AUTOMATED LIFECYCLE INFRASTRUCTURE MONITORING APPLICATION,* **Jongseong Choi***, Registered at Republic of Korea in Nov 2025 (10-2025-1068454)

[A7] *SPEED LOG DATA AND PHYSICAL KNOWLEDGE NEURAL NETWORK-BASED ELECTRIC VEHICLE DYNAMIC COEFFICIENT ESTIMATION AND ENERGY CONSUMPTION PREDICTION METHOD*, **Jongseong Choi***, filed at Republic of Korea in Feb 2025 (10-2025-0017752)

[A6] *GAUSSIAN SPLATTING-BASED LIDAR-RGB MULTI-SENSOR AND RENDERING SYSTEM FOR ULTRA-HIGH-RESOLUTION FACILITY SCANNING AND MODELING,* **Jongseong Choi***, filed at Republic of Korea in Feb 2025 (10-2025-0017753)

[A5] *A MIXED REALITY DEVICE VISUALIZING THE LIDAR SCANNING ALIGNMENT PROCESS IN THE REAL-WORLD ENVIRONMENT THROUGH AN MR HEADSET,* **Jongseong Choi***, filed at Republic of Korea in Feb 2025 (10-2021-0017754)

[A4] *3D GAUSSIAN SPLATTING DIGITAL TWIN PLATFORM,* **Jongseong Choi***, filed Apr 2025 (10-2025-0046922)

[A3] *DIGITAL TWIN GENERATION SYSTEM FOR VISION-BASED STRUCTURAL HEALTH MONITORING,* **Jongseong Choi***, filed at Republic of Korea in Apr 2025 (10-2025-0046923)

[A2] *LOG DATA-BASED ELECTRIC VEHICLE ENERGY CONSUMPTION ANALYSIS AND PATH EXPLORATION SYSTEM,* **Jongseong Choi***, filed at Republic of Korea in Apr 2025 (10-2025-0058452)

[A1] *DIGITAL TWIN-BASED CONSTRUCTION SITE MONITORING SYSTEM USING MOBILE APP PERFORMING LOCAL UPDATE ON MAPPING API,* **Jongseong Choi***, filed at Republic of Korea in Apr 2025 (10-2025-0058453)

BOOK CHAPTER

[B1] Changwoon Han and **Jongseong Choi** (2022.10). Chapter 10: Information of PHM research and facilities, *PHM BOK Guide: Prognostics and Health Management Body of Knowledge* (pp. 307-345). Hongreung Publishing. ISBN: 979-11-5600-966-5 **(2023 Sejong's Choice Selection)**

TOP AI CONFERENCES (80 published; *corresponding author)

[W5] (accepted)Minhyeok Lim and **Jongseong Choi***, "Degree-Equalized View-Graph Selection under Tight Pair Budgets", IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV 2027**), January 6 – 8, 2027

[W4] (accepted)Sooyeun Yang and **Jongseong Choi***, "METRIC-GS: Selective Metric-Scale Estimation for 3D Gaussian Splatting", IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV 2027**), January 6 – 8, 2027

[W3] **Jongseong Choi***, "Micro-Splatting: Neural Rendering for Sim-to-Real Transfer in Physical AI and Robot Foundation Models", Nature Conference: Autonomous Robotics, Daejeon, South Korea, Oct 14 – 15, 2026

[W2] Semin Bae, Sungwook Choi, Chul Min Yeum, **Jongseong Choi***, "Physically Interactable Gaussian Splatting for Shared MR–VR Remote Inspection", IEEE International Symposium on Mixed and Augmented Reality (**IEEE ISMAR2026**), Bali, Italy, Oct 7 – 9, 2026

[W1] Hansol Lim, Minhyeok Im, and **Jongseong Choi***, "PRISM: Color-Stratified Point Cloud Sampling", The 28th International Conference on Pattern Recognition (**ICPR 2026**), Lyon, France, Aug 17 – 22, 2026

PEER-REVIEWED JOURNAL PAPERS (27 published, 1 accepted, 4 under review); *corresponding author

[J32] Jeewon Lee and **Jongseong Choi***, LiteVoxel: Low-memory Intelligent Thresholding for Efficient Voxel Rasterization, *ACM Transactions on Graphics* (IF: 9.5, JCR top 3.87%); *Under Review, the first submission date of 7th Nov, 2025

[J31] Hansol Lim, **Jongseong Choi***, Jeewon Lee, Haeseong Jeoung, and Minkyu Han, A Hybrid Surrogate for Electric Vehicle Parameter Estimation and Power Consumption via Physics-Informed Neural Operators, *Engineering Applications of Artificial Intelligence* (IF: 8.0, JCR top 8.99%); *Under Review, completed 2nd-round review with the first submission date of 31th Oct, 2025

[J30] Hansol Lim, Jeewon Lee, Jonathan Boyack, and **Jongseong Choi***, EV-PINN: Electric Vehicle Battery Surrogate Model Using Transformer-Based Physics-Informed Neural Network, *IEEE Transactions on Intelligent Vehicles* (IF: 14.3, JCR top 1.90%); *Under Review, the first submission date of 21st Mar, 2025

[J29] Jonathan Boyack, **Jongseong Choi***, Hansol Lim, Sangho Song, Jaecheul Park, and Hwasup Jang (2026), Enhancing Visual Data Localization and Extraction by Hyper-Realistic Scene Reconstruction with 3D Gaussian Splatting: Validation on a Full-Scale Ship Engine, *Computers in Industry* (IF: 10.0, JCR top 9.73%); *Under Review, completed 3rd-round review with the first submission date of 7th Apr, 2025

[J28] Hansol Lim, **Jongseong Choi***, Jee Won Lee, Jonathan Boyack, Seungchul Lee, and Hyungchai Park (2027), EV-PINN: Physics-Informed Neural Networks for Parameter Identification of Electric Vehicle Dynamics, *IEEE Transactions on Intelligent Vehicles* (IF: 14.3, JCR top 1.9%), accepted

[J27] Huaiyuan Weng, Syed Muhammad Rizvi, Chul Min Yeum*, **Jongseong Choi**, and Su-Min Kim (2027), RPGO-MR: Relocalization Pose Graph Optimization for accurate and latency-aware Mixed Reality localization, *Advanced Engineering Informatics*, 77, 105346 (IF: 11.5, JCR top 2.25%); doi: 10.1016/j.aei.2026.105346

[J26] Anas Alsharo, Max Midwinter, Chul Min Yeum*, and **Jongseong Choi** (2026), Generalizable UAV-based image registration framework for enhancing visual inspection of civil infrastructure, *ISPRS Journal of Photogrammetry and Remote Sensing*, vol 237, pp. 652-667 (IF: 12.9, JCR top 1.41%) doi: 10.1016/j.isprsjprs.2026.04.053

[J25] Jeewon Lee and **Jongseong Choi*** (2026), Micro-Splatting: Multistage Isotropy-informed Covariance Regularization Optimization for High-Fidelity 3D Gaussian Splatting, *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 1-12, (IF: 6.8, JCR top 7.81%); doi: 10.1109/TVCG.2026.3677074

[J24] Hanbeom Chang, **Jongseong Choi***, Chul Min Yeum (2026), 3D Reconstruction by Looking: Instantaneous Blind Spot Detection for Indoor SLAM through Mixed Reality, *Advanced Engineering Informatics,* 69, 104065 (IF: 11.5, JCR top 2.25%); doi: 10.1016/j.aei.2025.104065

[J23] Hansol Lim, Hanbeom Chang, **Jongseong Choi***, Chul Min Yeum (2025), LiDAR-3DGS: LiDAR Reinforcement for multimodal initialization of 3D Gaussian Splats, *Computers & Graphics*, 104293. doi: 10.1016/j.cag.2025.104293

[J22] Jonathan Boyack and **Jongseong Choi*** (2025), Photogrammetry Engaged Automated Image Labeling Approach, *Visual Informatics*, 100239. (IF: 4.4, JCR top 25.0%); doi: 10.1016/j.visinf.2025.100239

[J21] Francisco Yumbla*, Marcelo Fajardo, Anthony Piguave, Diego Ronquillo, Ricardo Ortiz, **Jongseong Choi**, Gabriel Diaz, and Xicu Garcia (2025), An Open-Source Multi-Robot Framework System for Collaborative Environments Based on ROS2, *IEEE Access*, vol. 13, pp. 16288-16302. doi: 10.1109/ACCESS.2025.3530391

[J20] Jonathan Boyack, **Jongseong Choi***, Jongryeol Jeong, Hyungchai Park, and Sewhan Kim (2024), LogPath: Log Data based Energy Consumption Analysis enabling Electric Vehicle Path Optimization, *Transportation Research Part D: Transport and Environment*, 135, 104387. (IF: 8.7, JCR top 7.58%); doi: 10.1016/j.trd.2024.104387

[J19] Lissette Iturburu, Xiaoyu Liu, Xin Zhang, Benjamin E. Wogen, Juan Nicolas Villamizar, Shirley J. Dyke, Julio Ramirez, **Jongseong Choi*,** and Gianella Valencia, Sergio M. Alcocer (2024), Building Pose Detection for the Characterization of Reinforced Concrete Buildings, *The Structural Design of Tall and Special Buildings, 33(13), e2020,* doi: 10.1002/tal.2120

[J18] Benjamin E. Wogen, **Jongseong Choi***, Xin Zhang, Xiaoyu Liu, Lissette Iturburu, and Shirly J. Dyke (2024), Automated Bridge Inspection Image Retrieval Based on Deep Similarity Learning and GPS. *ASCE Journal of Structural Engineering*, 150(3), 04023238. (IF: 3.8, JCR top 25.0%); doi: 10.1061/JSENDH.STENG-12639

[J17] Ok-Hue Cho and **Jongseong Choi*** (2024), A Comparative Analysis of IoT based Network Anomaly Detection and Prediction Using Vector Autoregressive Models, *Journal of Machine and Computing*, 127-137. doi: 10.53759/7669/jmc202404013.

[J16] Alfredo Valenzuela, **Jongseong Choi***, Ricardo Ortiz, Byungkon Kang, Michael Kim, and Taewook Kang (2023), Development of Mobile App to Enable Local Update on Mapping API: Construction Sites Monitoring Through Digital Twin, *Electronics*, 12(23), 4738. doi: 10.3390/electronics12234738

[J15] Xiao-Le Han, Toshiro Hata, **Jongseong Choi**, Yan-Jun Du, Yi-Jie Wang, & Ning-Jun Jiang* (2023), Deep Learning Based Approach for Automated Characterization of Large Marine Microplastic Particles, *Marine Environmental Research*, 183, 105829. doi: 10.1016/j.marenvres.2022.105829

[J14] Xiaoyu Liu, Shirley J. Dyke*, Ali Lenjani, Ilias Bilionis, Xin Zhang, & **Jongseong Choi** (2023), Automated Image Localization to Support Rapid Building Reconnaissance in a Large-scale Area, *Computer-Aided Civil and Infrastructure Engineering*, 38(1), 3-25. doi: 10.1111/mice.12828 (IF: 9.2, JCR top 3.63%)

[J13] Ju An Park, Xiaoyu Liu, Chul Min Yeum*, Shirley J. Dyke, Max Midwinter, Chungwook Sim**, Jongseong Choi**, Zhiwei Chu, Thomas Hacker, & Bedrich Benes (2022), Multioutput Image Classification to Support Post-Earthquake Reconnaissance, *ASCE Journal of Performance of Constructed Facilities*, 36(6), 04022063. doi: 10.1061/(ASCE)CF.1943-5509.0001755 **(Editor's Choice Selection)**

[J12] Xiao-Le Han, Ning-Jun Jiang*, Yu-Fei Yang, **Jongseong Choi**, Devandra N. Singh, Yan-Jun Du, & Yi-Jie Wang (2022), Deep Learning Approach for the Detection and Instance Segmentation of Clayey Soil Desiccation Crack, *Computers and Geotechnics*, 146, 104733. (IF: 7.1, JCR top 7.34%); doi: 10.1016/j.compgeo.2022.104733

[J11] **Jongseong Choi***, Lazaros Toumanidis, Chul Min Yeum, Patrikakis Charalampos, Ali Lenjani, Xiaoyu Liu, Panagiotis Kasnesis, Ricardo Ortiz, Nin-Jun Jiang, & Shirley J. Dyke (2022), Automated Graffiti Detection: Automated Graffiti

Detection: A Novel Approach for Maintaining Historical Architectures in Community, *Applied Sciences*, 12(6), 2983. doi: 10.3390/app12062983

[J10] **Jongseong Choi***, Ju An Park, Shirley J. Dyke, Chul Min Yeum, Xiaoyu Liu, Ali Lenjani, & Ilias Bilionis (2022), Similarity Learning to Enable Buliding Searches in Post-event Image Data, *Computer-Aided Civil and Infrastructure Engineering,* 37(2), 261-275. (IF: 9.2, JCR top 3.63%); doi: 10.1111/mice.12698

[J9] Xiaoyu Liu, Shirley J. Dyke*, Chul Min Yeum, Ilias Bilionis, Ali Lenjani, & **Jongseong Choi** (2020), Automated Indoor Image Localization to Support Post-Event Building Assessment. *Sensors*, 20(6), 1610. doi: 10.3390/s20061610

[J8] **Jongseong Choi*** & Shirley J. Dyke (2020), CrowdLIM: Crowdsourcing to Enable Lifecycle Infrastructure Management. *Computers in Industry*, 115, 103185. (IF: 10.0, JCR top 9.73%); doi: 10.1016/j.compind.2019.103185

[J7] Ali Lenjani*, Shirley J. Dyke, Ilias Bilionis, Chul Min Yeum, Kenzo Kamiya, **Jongseong Choi**, Xiaoyu Liu, & Arindam G. Chowdhury (2020), Towards Fully Automated Post-event Data Collection and Analysis: Pre-event and Post-event Information Fusion. *Engineering Structure,* 109884. (IF: 7.6, JCR top 8.29%); doi: 10.1016/j.engstruct.2019.109884

[J6] Chul Min Yeum*, **Jongseong Choi,** & Shirley J. Dyke. (2019), Automated Region-of-interest Localization and Classification for Vision-based Visual Assessment of Civil Infrastructure. *Structural Health Monitoring*, 1475921718765419. (IF: 5.7, JCR top 12.92%); doi: 10.1088/1361-665X/aa510e

[J5] **Jongseong Choi**, Chul Min Yeum*, Shirley J. Dyke, & Mohammad J. Jahanshahi (2018), Computer-aided Approach for Rapid Post-event Visual Evaluation of a Building Façade. *Sensors*, 18(9), 3017. doi: 10.3390/s18093017

[J4] Chul Min Yeum*, **Jongseong Choi**, & Shirley J. Dyke (2017), Autonomous Image Localization for Visual Inspection of Civil Infrastructure. *Smart Materials and Structures*, 26(3), 035051. doi: 10.1088/1361-665X/aa510e

[J3] Jeffrey A. Roux*, **Jongseong Choi**, & Neerad Shakya (2014), Parametric Scramjet Cycle Analysis for Nonideal Mass Flow Rate. *AIAA Journal of Thermophysics and Heat Transfer*, 28(1), 166-171. doi: 10.2514/1.T4217

[J2] Jeffrey A. Roux*, Neerad Shakya, & **Jongseong Choi** (2013**),** Scramjet: Minimum Thrust-specific Fuel Consumption with Material Limit. *AIAA Journal of Thermophysics and Heat Transfer*, 27(2), 367-368. doi: 10.2514/1.T4045

[J1] Jeffrey A. Roux*, Neerad Shakya, & **Jongseong Choi** (2012), Revised Parametric Ideal Scramjet Cycle Analysis. *AIAA Journal of Thermophysics and Heat Transfer,* 27(1), 178-183. doi: 10.2514/1.T3961

CONFERENCE PROCEEDINGS & POSTERS (86 published; *corresponding author)

[C86] Sooyeon Yang, Seungbok Choi*, **Jongseong Choi**, "The Effectiveness of Sliding Mode Control for Robust Vision Stabilization under the Plant-Binding Condition" The 26th International Conference on Control, Automation, and System (ICCAS 2026), Sapporo, Japan, October 27 – 30, 2026

[C85] Cheyul Lim and **Jongseong Choi***, "A Small Language Model Agent for Automated Review and Correction of Construction Administrative Documents", Annual Conference of Korean Society of Civil Engineers (KSCE 2026), Gangwon, South Korea

[C84] Hansol Lim and **Jongseong Choi***, "Harness Engineering Vision-Language Model Agent for Accurate Construction-Site BIM Information Retrieval", Annual Conference of Korean Society of Civil Engineers (KSCE 2026), Gangwon, South Korea

[C83] **Jongseong Choi*** et. al., "Micro-Splatting for Physical AI: High-Fidelity Neural Rendering, Digital Twins, and Applications to Automated Inspection of Nuclear Power Plants", The 16th International Conference on Quality, Reliability, Risk, Maintenance, and Safety Engineering (QR2MSE 2026), Qingdao, July 22 – 25, 2026.

[C76 - 82] Hansol Lim and **Jongeong Choi*,** "Nondimensionalized Physics-Informed Neural Operators for PHM Surrogate Modelling", Annual Conference of the PHM Korea (PHM Korea 2026), Westin Josun Hotel, Busan, South Korea, Jun 24 – 27, 2026. Plus, 6 students generated papers; **3 Best Paper Awards**

[C75] **(Invited) Jongseong Choi***, “Micro-Splatting: Manufacturing AI Neural Rendering Techniques”, 2026 Spring Conference of The Korean Society for Precision Engineering, Jeju Shinhwa World, Jeju, South Korea, May 13 – 15, 2026.

[C74] **(Invited) Jongseong Choi***, “Micro-Splatting: Manufacturing AI Neural Rendering Techniques for Addressing the Sim2Real Problem in Physical AI by Reflecting Material and Surface Properties”, 2026 Spring Conference of The Korean Society for Composite Materials, Jeju Shinhwa World, Jeju, South Korea, Apr 22 – 24, 2026.

[C64 - 73] **(Invited) Jongseong Choi*** and Jeewon Lee, “Neural Rendering Techniques for Manufacturing AI to Address the Sim2Real Problem in Physical AI”, Korean Society of Mechanical Engineers (KSME)-Reliability Division 2026, Lotte Hotel Jeju, Jeju, South Korea, Apr 8 – 10, 2026. Plus, 9 students generated papers; **1 Best Paper Award**

[C63] Sooyuen Yang, Hansol Lim, Jeewon Lee, and **Jongseong Choi***, “Resolving the Accuracy-Efficiency Trade-off: A Hierarchical Learning-Based Control Architecture”, The 25th International Conference on Control, Automation, and System (ICCAS 2025), Songdo Convensia, Incheon, South Korea, IEEE Xplore, Nov 4 – 7, 2025.

[C62] Hansol Lim, Jeewon Lee, Sooyeun Yang, and **Jongseong Choi***, “Small Lagrangian Networks for Nonlinear Model Predictive Control”, The 25th International Conference on Control, Automation, and System (ICCAS 2025), Songdo Convensia, Incheon, South Korea, IEEE Xplore, Nov 4 – 7, 2025.

[C61] Jeewon Lee, Hansol Lim, Sooyeun Yang, and **Jongseong Choi***, “Hybrid Vision Servoing with Deep Alignment and GRU-Based Occlusion Recovery”, The 25th International Conference on Control, Automation, and System (ICCAS 2025), Songdo Convensia, Incheon, South Korea, IEEE Xplore, Nov 4 – 7, 2025.

[C60] **(Invited) (Best Paper Award) Jongseong Choi***, “Visual Intelligence for Predictive Engineering System Assessment”, The 15th International Conference on Quality, Reliability, Risk, Maintenance, and Safety Engineering (QR2MSE 2025), Hohhot Juva Grand Hotel, Hohhot, China, Jul 23 – 26, 2025.

[C59] **(Invited) (Best Presentation Award) Jongseong Choi***, “Physics-Informed Neural Network (PINN) and Hyper-Realistic Digital Twin (3DGS) for Predictive Engineering System Assessment”, International Conference on Precision Engineering and Sustainable Manufacturing (PRESM 2025), Chiang Mai Marriott Hotel, Chiang Mai, Thailand, Jul 6 – 11, 2025.

[C58] **(Keynote) Jongseong Choi***, “Physics-Informed Neural Network (PINN) and Hyper-Realistic Digital Twin (3DGS) for Predictive Engineering System Assessment”, Annual Conference of Drive & Control, KIST Gangneung, Gangneung, South Korea, Jun 26 – 27, 2025.

[C57] Hanbeom Chang, Hyeji Chang, Sungwook Choi, and **Jongseong Choi***, “PHM-Driven XR-Enabled Sensor Fusion Platform for Remote Structural Health Management and Collaborative Inspection”, Annual Conference of the PHM Korea (PHM Korea 2025), Jeju Booyoung Hotel, Seoguipo, South Korea, Jun 23 – 25, 2025.

[C56] Alfredo Valenzuela, Hugo Zuniga, Felipe Zuniga, and **Jongseong Choi***, “ROS-Based PTZ Camera System for Visual Monitoring in PHM”, Annual Conference of the PHM Korea (PHM Korea 2025), Jeju Booyoung Hotel, Seoguipo, South Korea, Jun 23 – 25, 2025.

[C55] Sooyeun Yang and **Jongseong Choi***, “An Integrated Framework for Object Reconstruction with 3D Gaussian Splatting and Reinforcement-Learned Sliding-Mode Landing Control”, Annual Conference of the PHM Korea (PHM Korea 2025), Jeju Booyoung Hotel, Seoguipo, South Korea, Jun 23 – 25, 2025.

[C54] Jonathan Boyack, Hansol Lim, and **Jongseong Choi***, “An Integrated Framework for Object Reconstruction with 3D Gaussian Splatting and Reinforcement-Learned Sliding-Mode Landing Control”, Annual Conference of the PHM Korea (PHM Korea 2025), Jeju Booyoung Hotel, Seoguipo, South Korea, Jun 23 – 25, 2025.

[C53] Jeewon Lee and **Jongseong Choi***, “Linear Optimization Strategy for Individual Battery Cell Power Controller to Extend its Remaining Useful Life using PHM”, Annual Conference of the PHM Korea (PHM Korea 2025), Jeju Booyoung Hotel, Seoguipo, South Korea, Jun 23 – 25, 2025.

[C52] Hansol Lim and **Jongseong Choi***, “Interpretable Surrogates via Lagrangian Energy Heads”, Annual Conference of the PHM Korea (PHM Korea 2025), Jeju Booyoung Hotel, Seoguipo, South Korea, Jun 23 – 25, 2025.

[C51] Sooyeun Yang, Jeewon Lee, Hansol Lim, **Jongseong Choi***, "OPTIMIS: Omniverse-Powered 3D Gaussian Splatting", Korean Society of Mechanical Engineers (KSME)-Reliability Division 2025, Kolon Hotel, Gyeongju, South Korea, Apr 9 – 11, 2025.

[C50] Hansol Lim, Jeewon Lee, Sooyeun Yang, and **Jongseong Choi***, "Physics-Informed Diffusion Models for 2D Heat Equation", Korean Society of Mechanical Engineers (KSME)-Reliability Division 2025, Kolon Hotel, Gyeongju, South Korea, Apr 9 – 11, 2025.

[C49] Alfredo Valenzuela, **Jongseong Choi***, Felipe Zuniga, and Hugo Zuniga, "Visual Monitoring Assessments System through Robot Operating System (ROS)", Korean Society of Mechanical Engineers (KSME)-Reliability Division 2025, Kolon Hotel, Gyeongju, South Korea, Apr 9 – 11, 2025.

[C48] Cheyul Im, Hansol Lim, and **Jongseong Choi***, "Virtual Reality based Remote Controlled Perspective Synchronized Snowplow", Korean Society of Mechanical Engineers (KSME)-Reliability Division 2025, Kolon Hotel, Gyeongju, South Korea, Apr 9 – 11, 2025.

[C47] Sungwook Choi, Hanbeom Chang, **Jongseong Choi***, "Remote Collaboration and Real-Time Visualization System Using Mixed Reality: Application in Construction and Industrial Sites", Korean Society of Mechanical Engineers (KSME)-Reliability Division 2025, Kolon Hotel, Gyeongju, South Korea, Apr 9 – 11, 2025.

[C46] Jeewon Lee, Hansol Lim, Sooyeun Yang, **Jongseong Choi***, Sang Hyuk Lee, Dongsoo Kang, Hankwang Choi, and Yongkyu Kim, "NuClearView: Object-Centric 3D Reconstruction and Automated Defect Localization for Nuclear Reactor Inspection", Korean Society of Mechanical Engineers (KSME)-Reliability Division 2025, Kolon Hotel, Gyeongju, South Korea, Apr 9 – 11, 2025.

[C45] Lesus Davila, Alfredo Valenzuela, Ricardo Ortiz, **Jongseong Choi**, Francisco Yumbla*, "Software Architecture and Simulation Interface for Autonomous Underwater Vehicles", IEEE Ecuador Technical Chapters Meeting 2024 (ETCM 2024), Universidad de Cuenca, Cuenca, Ecuador, Oct 15 – 18, 2024

[C44] Jee Won Lee, Hansol Lim, **Jongseong Choi***, Sangho Song, Jaechul Park, Hwasup Jang, "Development of a 3D Gaussian Splatting Digital Twining Platform for Remote Ship Assessment", 7th International Conference on Materials and Reliability (ICMR 2024), The Westin Josun Busan, Busan, South Korea, Dec 3-6, 2024

[C43] Hansol Lim, Jonathan Boyack, **Jongseong Choi***, "Runge-Kutta Neural Networks as PINN Alternative", 7th International Conference on Materials and Reliability (ICMR 2024), The Westin Josun Busan, Busan, South Korea, Dec 3-6, 2024

[C42] Hansol Lim, Hanbeom Chang and **Jongseong Choi***, "Transforming SLAM Data into 3D Gaussian Splatting Models", 12th International Conference on Robot Intelligence Technology and Applications (RiTA2024), UNIST, Ulsan, South Korea, Dec 4 – 7, 2024.

[C41] Hanbeom Chang, Hansol Lim, and **Jongseong Choi***, "3D Reconstruction by Looking: Instantaneous Visualization of Point Cloud Registration on an MR Device", 12th International Conference on Robot Intelligence Technology and Applications (RiTA2024), UNIST, Ulsan, South Korea, Dec 4 – 7, 2024.

[C40] **(Grand Prize)** Hansol Lim, Jeewon Lee, Sooyeun Yang, and **Jongseong Choi***, "Real-time Digital Twin Registration System for Visual Inspection in Mixed Reality", Korean Society of Mechanical Engineers (KSME)-Chungcheong Division 2024, KIMM, Daejeon, South Korea, Nov 21, 2024.

[C39] Sooyeun Yang, Jeewon Lee, Hanbeom Chang, and **Jongseong Choi***, "PhysGaussian-Enhanced Digital Twin for Building Safety Assessment", Korean Society of Mechanical Engineers (KSME)-Chungcheong Division 2024, KIMM, Daejeon, South Korea, Nov 21, 2024.

[C38] Jeewon Lee, Hansol Lim, Hanbeom Chang, Sooyeun Yang, and **Jongseong Choi***, "Augmented Reality-Driven Remote Inspection: Head-Movement and Gesture-Based PTZ Camera Control in Real-Time", Korean Society of Mechanical Engineers (KSME)-Chungcheong Division 2024, KIMM, Daejeon, South Korea, Nov 21, 2024.

[C37] **(Best Poster Award)** Hanbeom Chang, Soo-Jung Chi, **Jongseong Choi***, and Chul Min Yeum, "LiDAR-Camera based Real-time 3D Reconstruction with Mixed Reality Visualization System", Annual Conference of the PHM Korea (PHM Korea 2023), The K Hotel, Seoul, South Korea, Jun 25 – 26, 2024.

[C36] Hansol Lim, Hanbeom Chang, and **Jongseong Choi***, "ChromaFilter: Color-Based LiDAR Filter for Real-Time Feature Extraction and Optimization", Annual Conference of the PHM Korea (PHM Korea 2023), The K Hotel, Seoul, South Korea, Jun 25 – 26, 2024.

[C35] Sooyeun Yang, Hojin Song, Eugene Pak, and **Jongseong Choi***, "CNN-Driven Pothole Detection and Road Hazard Localized DB Development", Annual Conference of the PHM Korea (PHM Korea 2023), The K Hotel, Seoul, South Korea, Jun 25 – 26, 2024.

[C34] Yoonseong Kim, Hansol Lim, and **Jongseong Choi***, "Photogrammetry-Based Approach to Indoor Space Digital Twin Model Generation", Annual Conference of the PHM Korea (PHM Korea 2023), The K Hotel, Seoul, South Korea, Jun 25 – 26, 2024.

[C33] **(Invited)** **Jongseong Choi***, "Visual Analytics and Human-Machine Collaborative Remote Assessment Approach for Infrastructure Management", 20th World Conference on Non-Destructive Testing 2024 (WCNDT 2024), Songdo Convensia, Incheon, South Korea, May 27 – 31, 2024.

[C32] Hansol Lim and **Jongseong Choi***, "NeRF Generated Digital Twin Model for Real-Time, Perspective-Free Visual Analytics", 20th World Conference on Non-Destructive Testing 2024 (WCNDT 2024), Songdo Convensia, Incheon, South Korea, May 27 – 31, 2024.

[C31] Hanbeom Chang, **Jongseong Choi***, Ricardo Ortiz, and Chul Min Yeum, "Digital Twin Platform for Remote Assessment: Application in Extensive Ship Engine Inspection", 20th World Conference on Non-Destructive Testing 2024 (WCNDT 2024), Songdo Convensia, Incheon, South Korea, May 27 – 31, 2024.

[C30] Alfredo Valenzuela, Jee Won Lee, Chul Min Yeum, Ricardo Ortiz, and **Jongseong Choi***, "Remote Monitoring Assessment Through Pan-Tilt-Zoom Automated Camera Control", 20th World Conference on Non-Destructive Testing 2024 (WCNDT 2024), Songdo Convensia, Incheon, South Korea, May 27 – 31, 2024.

[C29] Jonathan Boyack, & **Jongseong Choi***, "Data Localization on A Digital Twin Model to Enable Remote Assessment: Implementation on a full-scale Ship Engine", 20th World Conference on Non-Destructive Testing 2024 (WCNDT 2024), Songdo Convensia, Incheon, South Korea, May 27 – 31, 2024.

[C28] Ricardo Ortiz, Alfredo Valenzuela, and **Jongseong Choi***, "Enhanced Construction Site Monitoring with RTK-GPS Guided Unmanned Mobile Robot and Mapping API Integration", 20th World Conference on Non-Destructive Testing 2024 (WCNDT 2024), Songdo Convensia, Incheon, South Korea, May 27 – 31, 2024.

[C27] **(Outstanding Paper Award)** Hansol Lim and **Jongseong Choi***, Alfredo Valenzuela, and Hanbeom Chang, "Real-Time 3D Video Generation using NeRF", The 14th International Symposium on NDT in Aerospace (AeroNDT2023), The Westin Josun Busan, Busan, South Korea, Nov 5 – 8, 2023.

[C26] **(The Best Paper Award)** Jonathan Boyack, Hansol Lim, Alfredo Valenzuela, **Jongseong Choi***, Sangho Song, Yongseok Choi, and Dongguk Im, "Region of Interest Locator with Digital Twin Model Generation of Large-Scale Facility", The 14th International Symposium on NDT in Aerospace (AeroNDT2023), The Westin Josun Busan, Busan, South Korea, 5 – 8 Nov 2023.

[C25] Alfredo Valenzuela, Jee Won Lee, Chul Min Yeum, Ricardo Ortiz, and **Jongseong Choi***, "Automated Pan-Tilt-Zoom Camera Control to Enable Monitoring Assessment", The 14th International Symposium on NDT in Aerospace (AeroNDT2023), The Westin Josun Busan, Busan, South Korea, 5 – 8 Nov 2023.

[C24] **(Best Paper Award)** Jonathan Boyack, & **Jongseong Choi***, "Falling Hazard Assessment Application: Automated Labelling Large Data Sets for Transfer Learning Models", Korean Society of Mechanical Engineers (KSME) Annual Meeting 2023, Songdo Convensia, Incheon, South Korea, Nov 1 – 4, 2023.

[C23] Alfredo Valenzuela, Ricardo Ortiz, Sambridha Bhattarai, and **Jongseong Choi***, "Localize Map Update Mobile App to Enable Construction Sites Monitoring: User-Guided Path Generation for Robot Operation", Korean Society of Mechanical Engineers (KSME) Annual Meeting 2023, Songdo Convensia, Incheon, South Korea, Nov 1 – 4, 2023.

[C22] **(Excellent Presentation Award)** Hansol Lim and **Jongseong Choi***, "Seamless 3D Scene Control of NeRF Generated Digital Twin Environments", Korean Society of Mechanical Engineers (KSME) Annual Meeting 2023, Songdo Convensia, Incheon, South Korea, Nov 1 – 4, 2023.

[C21] Hanbeom Chang, Jaewon Lee, Danielle Macmaster, Leeroy Makusha, **Jongseong Choi***, and Chul Min Yeum, "Human-Machine Collaborative Platform in Metaverse", Korean Society of Mechanical Engineers (KSME) Annual Meeting 2023, Songdo Convensia, Incheon, South Korea, Nov 1 – 4, 2023.

[C20] Alfredo Valenzuela and **Jongseong Choi***, "Locally Update Scenes to Enable a User-Guided Path Planning", Annual Conference of the PHM Korea (PHM Korea 2023), The K Hotel, Seoul, South Korea, July 7 – 8, 2023.

[C29] **(Best Paper Award)** Hansol Lim and **Jongseong Choi***, "Real-Time, Non-Intrusive Inspection of Structures and Machines Using Visual Analytics and SfM-Based Digital Twin Generation", Annual Conference of the PHM Korea (PHM Korea 2023), The K Hotel, Seoul, South Korea, July 7 – 8, 2023.

[C18] Ricardo Ortiz, Alfredo Valenzuela & **Jongseong Choi***, "Development of RTK-GPS Engaged Autonomous Mobility: Implementation and Analysis for Structural Health Monitoring Application", Advanced Smart Materials and Structures 2nd International Conference (ASMaS2023), Ton Duc Thang University, Ho Chi Minh, Vietnam, Jun 28 – 30, 2023.

[C17] Jonathan Boyack, & **Jongseong Choi***, "An Automated Falling Hazard Assessment of High-Rise Buildings Using Transfer Learning and Orthophoto Generation", Advanced Smart Materials and Structures 2nd International Conference (ASMaS2023), Ton Duc Thang University, Ho Chi Minh, Vietnam, Jun 28 – 30, 2023.

[C16] **Jongseong Choi***, "Automating Visual Analytics to Aid Lifecycle Management of Infrastructure", Structural Congress 2023, Hyatt Regency New Orleans, New Orleans, LA, May 3 – 6, 2023.

[C15] Jeewon Lee, Hanbeom Chang, **Jongseong Choi***, and Chul Min Yeum, "Automated Pan-Tilt-Zoom Camera Control Enabling Long-Range Visual Exploration for Extraterrestrial Missions", Korean Society of Mechanical Engineers (KSME)-Reliability Division 2023, Jeju Oriental Hotel, Jeju, South Korea, Mar 22 - 24, 2023.

[C14] **(Best Paper Award)** Pureun Jeong, Hansol Lim, Jimin Shin, and **Jongseong Choi***, "SLAM for Skid-Steering Mobile Robot Systems", Korean Society of Mechanical Engineers (KSME)-Reliability Division 2023, Jeju Oriental Hotel, Jeju, South Korea, Mar 22 - 24, 2023.

[C13] Ricardo Ortiz, Jee Won Lee, Jonathan Boyack, & **Jongseong Choi***, "Development of User-Designated-Path Driving Ground Robot for Construction Sites", Korean Society of Mechanical Engineers (KSME) Annual Meeting 2022, Jeju International Convention Center, Seoguipo, South Korea, Nov 9 – 12, 2022.

[C12] Jee Won Lee, Chul Min Yeum, Ricardo Ortiz, & **Jongseong Choi***, "Automated Pan-Tilt-Zoom Camera Control to Enable Long-Range Visual Assessment and Localization", Korean Society of Mechanical Engineers (KSME) Annual Meeting 2022, Jeju International Convention Center, Seoguipo, South Korea, Nov 9 – 12, 2022.

[C11] **(Best Paper Award)** Jonathan Boyack & **Jongseong Choi***, "A Real-Time Log Data Analytics-based Electric Vehicle Path Optimization", Korean Society of Mechanical Engineers (KSME) Annual Meeting 2022, Jeju International Convention Center, Seoguipo, South Korea, Nov 9 – 12, 2022.

[C10] Jonathan Boyack & **Jongseong Choi***, "Electric Vehicle Path Optimization using Real-Time Log Data for Decision Making", Annual Conference of the PHM Korea, The K Hotel, Seoul, South Korea Jun 29 – July 1, 2022.

[C9] **Jongseong Choi** & Shirley J. Dyke*, "ARIO: Automatic Reconnaissance Image Organizer to Learn from Earthquake", Annual Conference of Korean Society of Mechanical Engineers, Division of Reliability, Jeju Oriental Hotel, Jeju, South Korea, Mar 23 – 25, 2022.

[C8] Shirley J. Dyke*, Xiaoyu Liu, **Jongseong Choi**, Chul Min Yeum, Juan Park, Ali Lenjani, Julio A. Ramirez, & Randall Poston, "Learning from Earthquakes Using the Automatic Reconnaissance Image Organizer," Proceedings of 17th World Conference on Earthquake Engineering, Sendai International Center, Sendai, Japan, Sep 27- Oct 2, 2021.

[C7] **Jongseong Choi*** & Jonathan Boyack, "Automating Visual Data Analytics to Aid lifecycle Management of Infrastructure toward Sustainable cities", Annual Conference of the PHM Korea, The K Hotel, Seoul, South Korea, Sep 8 – 10, 2021.

[C6] **Jongseong Choi***, Ju An Park, Chul Min Yeum, & Shirley J Dyke, "Similarity Learning to Building Search Capability: Post-event Image Data Application", Proceedings of Asia Pacific Conference of the Prognostics and Health Management Society 2021, Jeju Ramada Hotel, Jeju, South Korea, Sep 8 – 11, 2021.

[C5] Jonathan Boyack & **Jongseong Choi***, "Machine Learning Approach for a Rapid Falling Hazard Assessment on High-Rise Buildings", Proceedings of Asia Pacific Conference of the Prognostics and Health Management Society 2021, Jeju Ramada Hotel, Jeju, South Korea, Sep 8 – 11, 2021.

[C4] Audai Theinat, Anahita Modiriasari, Antonio Bobet, Jay Melosh, Shirley J. Dyke*, Julio A. Ramirez, **Jongseong Choi**, Amin Maghareh, & Daniel Gomez, "Geology Explorations of Lava Tubes in the National Beds Lava Monuments," 50th Lunar and Planetary Science Conference (LPSC 2019), Woodlands Waterway Marriott Hotel and Convention Center, Woodlands, Texas, USA, Mar 18 – 22, 2019.

[C3] **Jongseong Choi***, Chul Min Yeum, Shirley J. Dyke, Mohammad R. Jahanshahi, & Gun Wook Park, "Rapid Vision-Based Inspection of Nonstructural Components in Buildings" Proceedings of the 9th European Workshop on Structural Health Monitoring (EWSHM 2018), Hilton Manchester Deansgate, Manchester, UK, July 10-13, 2018.

[C2] Chul Min Yeum*, **Jongseong Choi**, & Shirley J. Dyke, "Automated Region-of-Interest Localization and Classification for Visual Assessment on Civil Infrastructure," Proceedings of the 11th International Workshop on Structural Health Monitoring, Stanford University, Stanford, CA, USA, Sep 12 – 14, 2017.

[C1] Chul Min Yeum*, **Jongseong Choi**, & Shirley J. Dyke, "Image Localization for Computer-enhanced Visual Inspection of Civil Infrastructure," Proceedings of Engineering Mechanics Institute Conference, Price Center at UCSD, San Diago, CA, USA, Jun 4 – 7, 2017

TECHNICAL REPORT & GUEST EXPERT COLUMN

[R3] **(Special Lecture Course) Jongseong Choi**, "AI to Hardware: Neural Rendering for Sim2Real", The Korean Society for Noise and Vibration Engineering (한국소음진동공학회), Feb 2026

[R2] **(Special Issue Manuscript) Jongseong Choi**, "Physical AI, Sim2Real, and Neural Rendering Techniques", The Institute of Electronics and Information Engineers (IEIE, 한국전자공학회), Feb 2026

[R1] Chul Min Yeum, Anup Mohan, Shirley J. Dyke, Mohammad Jahanshahi, **Jongseong Choi**, Ziyi Zhao, Ali Lenjani, and Julio A. Ramirez, "Image-based collection and measurements for construction pay items." *Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2017/10*. West Lafayette, IN: Purdue University, Mar 2015

PROPOSAL DEVELOPMENT to U.S. GOVERNMENT (5 generated, 1 awarded).

[P5] **Active Citizen Engagement to Enable Lifecycle Management of Infrastructure Systems** *07/2016*

- **(Funded** under Grant No. **CMMI-1645047)** Co-authored successful proposal with funded $100,000 from National Science Foundation (NSF).

[P4] **Automating Damage Quantification, Localization and BIM Updating Using Optical Data** *02/2020*

- Co-authored successful proposal and requested $400,000 to National Science Foundation (NSF).

[P3] **HDBE (E-Defense): Enabling Building Damage Assessment by Engaging Remote Experts** *01/2018*

- Co-authored and requested $700,000 to National Science Foundation (NSF).

[P3] S&SA: Autonomous Infrastructure Inspection and Condition-Based Maintenance *05/2017*

- Co-authored proposal and requested to National Science Foundation (NSF).

[P1] S&SA: Reconfigurable Aerial Robots for Intelligent Assessment to Industrial Disasters *11/2016*

- Co-authored proposal and requested to National Science Foundation (NSF).